MOHD AZEEM

Full Stack Software Engineer | Java / Spring Boot | Python / FastAPI | Node.js | React / Electron / Android | AI System Design

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SUMMARY

Full Stack Software Engineer building production systems since 2024 across **Java / Spring Boot** and **Python / FastAPI** microservices, a **Node.js / Socket.IO** real-time server, and **React / Electron** desktop and **Android** clients for **Ahoy**, a $100K^{+}$ MAU AI team-collaboration platform. Hands-on in **system design** for multi-tenant microservices and **AI system design** for RAG, LLM agents and evaluation; previously built payment and fraud services in fintech.

TECHNICAL SKILLS

Languages: Java (8–21), Kotlin, Python, TypeScript, JavaScript (ES6+), SQL, Go, C/C++

Backend & Real-time: Spring Boot 3, Spring Cloud, Nacos, Microservices, REST APIs, FastAPI, Flask, Node.js, Socket.IO, MyBatis-Flex, Dropwizard, JDBI, Liquibase, Feign Client

Frontend, Desktop & Mobile: React.js, Electron.js, Zustand, TanStack Query, Ant Design, Tailwind CSS, Android (Jetpack, MVVM, Coroutines, RxJava, Room, Retrofit/OkHttp)

AI / LLM: OpenAI API & Agents SDK, Gemini, Anthropic Claude, RAG, ChromaDB, LangChain, Embeddings, AI Agents, LLM Evaluation & Observability, Prompt Engineering

System Design: Distributed Systems, Multi-tenant Architecture, Event-Driven Design, Horizontal Scaling, Caching, AI System Design

Data & Infra: MySQL, PostgreSQL, MongoDB, Redis, SQLite/SQLCipher, AWS (EC2, S3, SQS, CloudWatch), Azure, Docker, Firebase, Sentry, CI/CD

Security & Testing: JWT, Role-Based Access Control, IDOR Remediation, Tenant Isolation, TestNG, RestAssured

EXPERIENCE

Backend Engineer (Full Stack) **May 2025 – Present**

Novum Studio Ltd | Ahoy AI Team-Collaboration Platform (Web / Desktop / Android / Backend) *London, UK (Remote)*

- Architected **Java 17 / Spring Boot 3** microservices for a **multi-tenant** platform with **Spring Cloud + Nacos**, MySQL (MyBatis-Flex, Liquibase), MongoDB, **Redis** and **AWS SQS/S3**; enforced **JWT role-based access control** and remediated a critical **IDOR** authentication bypass across 11 backend files.
- Designed the **Python AI backend**: **LLM agents** on the OpenAI Agents SDK with function tools, a multi-tenant **RAG** system over **ChromaDB** with per-workspace isolation and auto-indexing from Spring Boot, and FastAPI/Flask services on **Gemini**, plus **LLM tracing**, **cost metering** and an **evaluation harness that gates releases**.
- Shipped a human-in-the-loop **AI task-suggestion** workflow (LLM draft, then user confirm-create), improving task acceptance by **35%**, along with decision-summary and what's-new AI commands built on **OpenAI structured outputs**.
- Scaled a **Node.js / Socket.IO** real-time messaging server horizontally with Node cluster, sticky sessions and a **Redis adapter**; MongoDB message persistence, **SQS**-driven event delivery from the backend, and **Docker** deployments on AWS.
- Developed the cross-platform **Electron** desktop app (macOS/Windows) in **React + TypeScript** (Zustand, TanStack Query, Ant Design) with an encrypted **SQLite (SQLCipher)** local store, auto-updates and **Sentry** monitoring.
- Developed the **Android** app in **Kotlin/Java** (MVVM, Jetpack, Coroutines, RxJava, Retrofit/OkHttp) with offline-first **Room** storage and **Firebase** (FCM, Crashlytics, Analytics).

Backend Engineering Intern **January 2024 – December 2024**

ePayLater | Fintech, Buy-Now-Pay-Later Payments *Gurgaon, Haryana*

- Built **secure REST APIs** (Spring Boot, Dropwizard / JDBI) on **Microsoft Azure** processing **10K+ daily transactions**, cutting response time by **30%**; migrated services from **Java 8 to Java 21** with **virtual threads**, improving throughput by **15%**.
- Implemented **event-driven UPI fraud alerts** with **Redis pub/sub** for real-time threshold monitoring, and automated **KYC & fraud checks** by integrating Signzy, PiChain, Freshdesk and WhatsApp Business with retry/fallback.

PROJECTS

PhD Report Management System | Spring Boot, React (Vite), PostgreSQL, JWT, Apache POI, iText **April 2025**

- Full-stack academic workflow platform for Scholars, Coordinators and RAC Members with OTP registration, **JWT** role-based access, automated DOCX report generation and digitally signed PDF export.

EDUCATION

B.Tech, Computer Science & Engineering (CGPA 8.5, University Topper) **2019 – 2023**

Maulana Azad National Urdu University (NAAC A+) | Hyderabad, Telangana